

DEFORMATIONS OF CANONICAL PAIRS

TOMMASO DE FERNEX AND CHRISTOPHER D. HACON

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ABSTRACT. In this paper we prove that under mild hypotheses, the nef values of canonical pairs are deformation invariant and that the effective cones of curves of Fano varieties with terminal singularities are locally constant under deformation. One can interpret this result as a rigidity property of Fano varieties.

1. INTRODUCTION

According to Mori Theory, the geometry of a Fano varieties is governed by its cone of effective curves. It follows by a result of Wiśniewski [Wiś91] that the cone of effective curve of a smooth Fano manifold is locally constant under small deformations. One can interpret this result as a rigidity property of Fano manifolds.

Wiśniewski's result is of a topological nature, relying on the Hard Lefschetz Theorem and the fact that the family is topologically locally trivial. Using completely different methods, we investigate to which extent this property holds for deformations of singular Fano varieties. It is important to observe that, as soon as we allow singularities, the topology will in general change in the family.

Our main result can be summarized as follows (for a more precise formulation, see Theorem 5.9).

Theorem 1.1. *If $f: X \rightarrow T$ is a flat projective family of Fano varieties with terminal $\mathbb{Q}$ -factorial singularities, then the Picard number of the fibers is constant and the cone of effective curves is locally constant in the family.*

It might be worthwhile to emphasize that, in fact, the essential geometric information carried by each extremal ray of the cone of curves of a given fiber of f remains unchanged as the ray “deforms” in the family. Simple examples show that the condition on singularities we consider in Theorem 1.1 is optimal. As we shall see, there is a surprising connection between this problem and a generalization of Siu's theorem on the invariance of plurigenera.

The main purpose of this paper is twofold. First, we investigate various deformation properties of canonical pairs, especially in connection to questions related to the deformation invariance of singularities and extension properties of log pluricanonical forms. This will be addressed in the first part of the paper. Then we apply these properties to study the behavior in families of nef values and cones of effective curves.

The general setting considered throughout the paper consists of a flat family $f: X \rightarrow T$, where T is a smooth curve with a fixed point 0 , and the central fiber $X_0 = f^{-1}(0)$ is a

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normal variety. If D is a $\mathbb{Q}$ -divisor on X not containing X_0 in its support, then we denote by $D|_{X_0}$ its restriction to X_0 as a $\mathbb{Q}$ -divisor.

Theorem 1.1 follows from an other result (see Corollary 5.2), which roughly says that, given a flat projective family $f: X \rightarrow T$ of varieties X_t , if A is an ample divisor on X and D is an effective $\mathbb{Q}$ -divisor such that $(X_0, D|_{X_0})$ is a Kawamata log terminal pair with canonical singularities, then for t in a neighborhood of 0, the $\mathbb{Q}$ -divisor $K_{X_t} + D|_{X_t}$ is $\mathbb{Q}$ -Cartier and, under some additional mild hypotheses, the nef value

$$\tau_{A|_{X_t}}(K_{X_t} + D|_{X_t}) := \min \{ \lambda \in \mathbb{R} \mid K_{X_t} + D|_{X_t} + \lambda A|_{X_t} \text{ is nef} \}$$

of $K_{X_t} + D|_{X_t}$ with respect to $A|_{X_t}$ is independent of t .

When f is a smooth family and $D = 0$, this is the aforementioned result of Wiśniewski yielding Theorem 1.1 for smooth families of Fano manifolds. A simple example shows that, in the above results, the hypotheses on the singularities are sharp. In fact, the same example shows that the analogue of Theorem 1.1 does not hold, in general, for families of log Fano varieties.

The main task to prove invariance of nef values is to understand the general structure of Mori extremal contractions (over T) on the total family (cf. Theorem 5.1). If $\psi: X \rightarrow Z$ is such a contraction, and its restriction to the central fiber $\psi|_{X_0}$ is not an isomorphism, then one needs to show that the restriction to the generic fiber $\psi|_{X_\eta}$ is also nontrivial. The interesting case is when $\psi|_{X_0}$ is birational. To deal with this case, we rely on the notions of singularities of pairs and on suitable results on the extension of log pluricanonical forms.

The starting point is Siu's theorem on the deformation invariance of plurigenera (cf. [Siu98]), and the work of Kawamata and Nakayama on the deformation of canonical and terminal singularities. The question addressed in this paper is under which conditions can these results be extended to the more general setting of log pairs.

We know by a result of Kollár and Mori (cf. [KM92]) that if the central fiber X_0 is $\mathbb{Q}$ -factorial and has terminal singularities, then the total space X is $\mathbb{Q}$ -factorial in a neighborhood of X_0 , and combining this with results of [BCHM06], it follows that small deformations of Kawamata log terminal pairs with canonical singularities (resp. terminal singularities) remain canonical (resp. terminal) (cf. Section 3). We deduce in particular that, for families of pairs, the property of being a (weak) log Fano variety with canonical/terminal singularities is open (see Corollary 3.9).

It is then natural to ask if deformation invariance of plurigenera holds also in this context. Kawamata has shown that Siu's theorem can be extended to the case of varieties with canonical singularities when the general fiber is of general type. The case of canonical pairs turns out to be more subtle. By a result of Hacon and McKernan [HM08] (cf. Theorem 4.1), we know that if $f: X \rightarrow T$ is a flat projective morphism, D is an effective $\mathbb{Q}$ -divisor such that $(X_0, D|_{X_0})$ is a Kawamata log terminal pair with canonical singularities, D is big over T , and $\text{SBs}(K_X + D/T)$ contains no components of $\text{Supp}(D|_{X_0})$, then $h^0(\mathcal{O}_{X_t}(m(K_{X_t} + D|_{X_t})))$ is constant on a neighborhood of $0 \in T$.

It turns out that the hypothesis that $\text{SBs}(K_X + D/T)$ contains no components of $\text{Supp}(D|_{X_0})$ is a necessary one (cf. Remark 4.5). In Section 4 we illustrate an interesting example in which this hypothesis may be avoided (see Theorem 4.6). The interest in this variant of the result is that the condition on the stable base locus is replaced by certain conditions that are automatically satisfied for families of Fano varieties.

These results are the main tools used in this paper to study the behavior of nef values in families and to prove Theorem 1.1.

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2. NOTATION AND CONVENTIONS

Throughout this paper we work over the field of complex numbers $\mathbb{C}$. A *divisor* on a normal variety will be understood to be a Weil divisor. If D is a divisor on a normal variety X , we denote $H^0(D) = H^0(\mathcal{O}_X(D))$. For a proper morphism of varieties $f: X \rightarrow S$, we denote by $N^1(X/Z)$ the Néron–Severi space (with real coefficients) of X over Z , and by $N_1(X/Z)$ its dual space. A surjective morphism of varieties with positive relative dimension is said to be a *fibration*.

The *stable base locus* $\text{SBs}(D)$ of a $\mathbb{Q}$ -divisor D on a normal variety X is defined as the intersection $\bigcap_m \text{Bs}(|mD|)$ of the base loci of the linear systems $|mD|$ taken over all $m > 0$ sufficiently divisible. For a morphism $f: X \rightarrow Z$, we denote $\text{SBs}(D/Z) := \bigcap_H \text{SBs}(D + f^*H)$, where the intersection is taken over all ample divisors H on Z .

A *pair* (X, D) consists of a positive dimensional normal variety X and a $\mathbb{Q}$ -divisor D such that $K_X + D$ is $\mathbb{Q}$ -Cartier. A *log resolution* of a pair (X, D) is a proper birational morphism from a smooth variety $\mu: Y \rightarrow X$ with the property that the exceptional set $\text{Ex}(\mu)$ is a divisor and the set $\text{Ex}(\mu) \cup \text{Supp}(K_Y - \mu^*(K_X + D))$, where $\mu_*K_Y = K_X$, is a simple normal crossings divisor. Given a log resolution $\mu: Y \rightarrow X$ of a pair (X, D) , we write $K_Y + \Gamma = \mu^*(K_X + D)$ where $\mu_*K_Y = K_X$ and $\mu_*\Gamma = D$. If $\Gamma = \sum \gamma_i P_i$ where P_i are prime divisors, then the *discrepancy* of P_i over (X, D) is defined as $a_{P_i}(X, D) = -\gamma_i$. The pair (X, D) is *Kawamata log terminal* (resp., *log canonical*) if $a_{P_i}(X, D) > -1$ (resp., $a_{P_i}(X, D) \geq -1$) for any prime divisor P_i over X . The pair (X, D) is *canonical* (resp., *terminal*) if $a_{P_i}(X, D) \geq 0$ (resp., $a_{P_i}(X, D) > 0$) for any prime divisor P_i exceptional over X , and moreover $\lfloor (1 - \epsilon)D \rfloor = 0$ for all small $\epsilon > 0$ (this last condition is only relevant when $\dim X = 1$, as it is redundant otherwise; note that when $\dim X = 1$ the first conditions are empty, since there are no exceptional divisors at all).

A variety X is *Fano* if it is a positive dimensional projective variety with ample $\mathbb{Q}$ -Cartier canonical divisor K_X . A pair (X, D) is a *log Fano variety* (resp., a *weak log Fano variety*) if X is projective, (X, D) had Kawamata log terminal singularities, and $K_X + D$ is ample (resp., nef and big).

3. $\mathbb{Q}$ -FACTORIORITY AND DEFORMATION OF SINGULARITIES OF PAIRS

We start by recalling the following result of Kollár and Mori.

Proposition 3.1. *Let X be a normal variety and $S \subset X$ a Cartier divisor that is normal and S_3 . Let D be any Weil divisor on X whose support does not contain S . If $\mathcal{O}_S(D|_S)$ is Cartier and there is a closed subset $Z \subset X$ with $\text{codim}(Z \cap S, S) \geq 3$ such that D is Cartier on $U = X - Z$, then $\mathcal{O}_X(D)$ is Cartier on a neighborhood of S .*

For the convenience of the reader, we include the arguments of [KM92, 12.1.8].

Proof. Since the question is local, we may assume that $\mathcal{O}_X(S) \cong \mathcal{O}_X$, and hence $\mathcal{O}_U(S|_U) \cong \mathcal{O}_U$. Let $i: U \rightarrow X$ be the inclusion. Multiplication by the restriction of a section of $\mathcal{O}_X(S)$ defining S gives a short exact sequence on U

$$0 \rightarrow \mathcal{O}_U(D|_U) \cong \mathcal{O}_U((D - S)|_U) \rightarrow \mathcal{O}_U(D|_U) \rightarrow \mathcal{O}_{U \cap S}(D|_{U \cap S}) \rightarrow 0.$$

Pushing forward via i and using the fact that as S is S_3 , then $R^1i_*\mathcal{O}_{U \cap S}(D|_{U \cap S}) = H^2_{Z \cap S}(\mathcal{O}_{U \cap S}(D|_{U \cap S})) = 0$, we obtain an exact sequence

$$0 \rightarrow \mathcal{O}_X(D) \rightarrow \mathcal{O}_X(D) \rightarrow \mathcal{O}_S(D|_S) \rightarrow R^1i_*\mathcal{O}_U(D|_U) \rightarrow R^1i_*\mathcal{O}_U(D|_U) \rightarrow 0.$$

The map $R^1i_*\mathcal{O}_U(D|_U) \rightarrow R^1i_*\mathcal{O}_U(D|_U)$ is induced by multiplication by the equation of S . By Nakayama's Lemma, it follows that $R^1i_*\mathcal{O}_U(D|_U) = 0$ on a neighborhood of S . But then $\mathcal{O}_X(D) \rightarrow \mathcal{O}_S(D|_S)$ is surjective and the claim follows. $\square$

Corollary 3.2. *Let S be a normal irreducible Cartier divisor on a variety X , and suppose that S is $\mathbb{Q}$ -factorial with terminal singularities.*

Then X is $\mathbb{Q}$ -factorial in a neighborhood of S .

Remark 3.3. Alternatively, one can give a proof of this result using extension theorems on log pluricanonical forms. In fact, following this approach, one can also check that the same result holds if S is only assumed to be Cartier in codimension 2.¹

Remark 3.4. If we are working in a neighborhood of a normal Cartier divisor S on a variety X , then X is automatically normal. Indeed, since S is regular in codimension one, and is Cartier, X is regular in codimension one near S . Let $f: X' \rightarrow X$ be the normalization. Since, over a neighborhood of S , f is an isomorphism away from sets of codimension ≥ 2 , the divisor $\tilde{S} := f^*S$ is irreducible and reduced. Note that $f|_{\tilde{S}}: \tilde{S} \rightarrow S$ is a finite birational surjective morphism of varieties. Since S is normal, $f|_{\tilde{S}}$ is an isomorphism by Zariski's Main Lemma. This implies that f is an isomorphism over a neighborhood of S , and therefore X is normal near S .

Remark 3.5. The corollary fails in general if the singularities are canonical but not terminal. For an example, one can consider the family of quadrics of equation $\{xy + z^2 + t^2u^2 = 0\} \subset \mathbb{P}^3 \times \mathbb{A}^1$, where (x, y, z, u) are homogeneous coordinates on $\mathbb{P}^3$ and t is a parameter on $\mathbb{A}^1$.

Proposition 3.6. *On a normal variety X , let S be a normal Cartier divisor, and D be a divisor whose support does not contain S . Assume that $K_S + D|_S$ is $\mathbb{Q}$ -Cartier and $(S, D|_S)$ is a canonical pair.*

Then $K_X + S + D$ is $\mathbb{Q}$ -Cartier on a neighborhood of S . Moreover:

- (a) *If $[D|_S] = 0$, then $(X, S + D)$ is purely log terminal and (X, D) is canonical in a neighborhood of S . In particular, if $f: X \rightarrow T$ is a flat morphism and $S = X_0$ is the fiber over a point $0 \in T$, then $(X_t, D|_{X_t})$ is canonical for every t in a neighborhood of 0 .*
- (b) *If $(S, D|_S)$ is terminal and $[D|_S] = 0$, then (X, D) is terminal in a neighborhood of S . In particular, if $f: X \rightarrow T$ is a flat morphism and $S = X_0$ for some $0 \in T$, then $(X_t, D|_{X_t})$ is canonical for every t in a neighborhood of 0 .*
- (c) *If S is $\mathbb{Q}$ -Gorenstein (equivalently, if $D|_S$ is $\mathbb{Q}$ -Cartier), then $(X, S + D)$ is log canonical in a neighborhood of S .*

Proof. Since $(S, D|_S)$ is canonical, S is Cohen–Macaulay. The case in which $\dim S = 1$ is trivial and the one in which $\dim S = 2$ is well known. We therefore assume that $\dim S \geq 3$ and hence S is S_3 . By the 2-dimensional case, it follows that (X, D) is canonical in codimension 2 near S . Therefore the closed subset $Z \subset X$ on which $K_X + S + D$ is not

¹Right?

$\mathbb{Q}$ -Cartier has codimension at least 3. By Proposition 3.1, $K_X + S + D$ is $\mathbb{Q}$ -Cartier on a neighborhood of X .

If $\lfloor D|_S \rfloor = 0$, then $(S, D|_S)$ is Kawamata log terminal, and since we have now established that $K_X + S + D$ is $\mathbb{Q}$ -Cartier, it is well known that $(X, S + D)$ is purely log terminal in a neighborhood of S (cf. [KM98, Theorem 5.50]). By the arguments of [BCHM06, Corollary 1.4.3] it follows that (X, D) is canonical in a neighborhood of S .

Assume now that $(S, D|_S)$ is terminal and $\lfloor D|_S \rfloor = 0$, and suppose that (X, D) is not terminal near S . Then there is an exceptional divisor over X whose discrepancy over (X, D) is ≤ 0 and whose center $W \subset X$ intersects X_0 . We can find an effective Cartier divisor $H \subset X$ containing W and intersecting properly X_0 . Note that the pair $(X, D + \epsilon H)$ is not canonical at W for any $\epsilon > 0$. On the other hand, if ϵ is sufficiently small, then $(X_0, D_{X_0} + \epsilon H_{X_0})$ is canonical and $\lfloor D_{X_0} + \epsilon H_{X_0} \rfloor = 0$. This contradicts (a).

Regarding the assertion in (c), if S is $\mathbb{Q}$ -Gorenstein, then $K_S + (1 - \epsilon)D|_S$ is $\mathbb{Q}$ -Cartier and $(S, (1 - \epsilon)D|_S)$ is Kawamata log terminal for every rational number $\epsilon > 0$, so that we conclude that $(X, S + (1 - \epsilon)D)$ is purely log terminal in a neighborhood of S and hence $(X, S + D)$ is log canonical on this neighborhood. $\square$

Remark 3.7. The example [Kaw99b, Example 4.3] shows that this result fails for Kawamata log terminal singularities.

Remark 3.8. For a pair (X, D) with canonical singularities, the property that $\lfloor D \rfloor \leq 0$ is equivalent to the pair being Kawamata log terminal.

Corollary 3.9. *Let $f: X \rightarrow T$ be a flat projective fibration from a normal variety to a smooth curve. Fix a point $0 \in T$, and suppose that the fiber X_0 over 0 is a normal variety. Assume that for some effective divisor D on X not containing X_0 in its support, the pair $(X_0, D|_{X_0})$ is a log Fano variety (resp. a weak log Fano variety) with canonical/terminal singularities.*

Then $(X_t, D|_{X_t})$ is a log Fano variety (resp. a weak log Fano variety) with canonical/terminal singularities for all t in a neighborhood of $0 \in T$.

Proof. As $(X_0, D|_{X_0})$ is a log Fano variety (resp. a weak log Fano variety), we have that $\lfloor D|_{X_0} \rfloor = 0$. Note that, since X_0 is a connected and reduced fiber, f has connected fibers. Since $(X_0, D|_{X_0})$ is canonical, the $\mathbb{Q}$ -divisor $K_X + D$ is $\mathbb{Q}$ -Cartier in a neighborhood of X_0 by Proposition 3.6, and the pair $(X_t, D|_{X_t})$ is canonical/terminal for all t in a neighborhood of 0 . Notice also that $\lfloor D \rfloor = 0$ in a neighborhood of X_0 , and hence $\lfloor D|_{X_t} \rfloor$ for t in a neighborhood of 0 . If $(X_0, D|_{X_0})$ is log Fano, then $-(K_{X_t} + D|_{X_t})$ is ample for every t in a neighborhood of $0 \in T$ since ampleness is open in families, and hence $(X_t, D|_{X_t})$ is log Fano for any such t . Assuming that $(X_0, D|_{X_0})$ is only weak log Fano, we conclude by the lemma that follows. $\square$

Lemma 3.10. *Let (X, D) be a Kawamata log terminal pair (with D an effective $\mathbb{Q}$ -divisor). Suppose that $f: X \rightarrow T$ is a flat projective fibration to a smooth curve, and let $0 \in T$ be a point such that the fiber X_0 is a normal variety, and is not contained in the support of D . Let L be a Cartier divisor on X such that $L|_{X_0}$ is nef and big and so is $aL|_{X_0} - (K_X + D)|_{X_0}$ for some $a \geq 0$. Then $L|_{X_t}$ is nef and big for all t in a neighborhood of $0 \in T$.*

Proof. Let $M \in \{L, raL - r(K_X + D)\}$, where r is the Cartier index of $aL - (K_X + D)$ (we can assume that a is rational). Note that $M|_{X_0}$ is nef and big. Then $M|_{X_t}$ is nef if $t \in T$ is very general (see [Laz04, Proposition 1.4.14]).

We claim that $M|_{X_t}$ is also big for very general $t \in T$. Indeed, since $M|_{X_0}$ is nef and big, all the higher cohomology groups $H^i(kM|_{X_0})$ fail to grow maximally as $k \rightarrow \infty$; more precisely, there is a constant C such that $h^i(\mathcal{O}_{X_0}(kM|_{X_0})) \leq Ck^{d-1}$ for every $i > 0$ and $k \geq 1$ where d is the dimension of X_0 . Note that f is equidimensional. By semicontinuity of cohomological dimensions [Har77, III.12.8], for every k there is a open neighborhood $U_k \subseteq T$ of 0 such that $h^i(\mathcal{O}_{X_t}(kM|_{X_t})) \leq Ck^{d-1}$ for every $i > 0$ and $t \in U_k$. We conclude that $h^i(\mathcal{O}_{X_t}(kM|_{X_t}))$ fails to grow maximally for very general t . On the other hand, note that $\mathcal{O}_X(kM)$ is flat over T (cf. [Har77, III.9.2(e) and III.9.2(c)]), and thus the Euler characteristic of $\mathcal{O}_{X_t}(kM|_{X_t})$ is constant as a function of t . We conclude that $h^0(\mathcal{O}_{X_t}(kM|_{X_t}))$ grows maximally (i.e., to the order of k^d) for very general t . This shows that $M|_{X_t}$ is big for very general $t \in T$.

So there is a set $W \subseteq T$, which is the complement of a countable set, such that $(X_t, D|_{X_t})$ is Kawamata log terminal and both $L|_{X_t}$ and $aL|_{X_t} - (K_X + D)|_{X_t}$ are nef and big for every $t \in W$. By [Kol93, Theorem 1.1], there is a positive integer m (depending only on d and a) such that $|mL|_{X_t}|$ is base point free for every $t \in W$, and for any such t the associated morphism $\phi_{|mL|_{X_t}|}$ is generically finite since $L|_{X_t}$ is big. By further shrinking W if necessary, we can also assume that the restriction map $H^0(\mathcal{O}_X(mL)) \rightarrow H^0(\mathcal{O}_{X_t}(mL))$ is surjective (cf. [Har77, III.12.8 and III.12.9]). This implies that $\phi_{|mL|_{X_t}|} = \phi_{|mL|}|_{X_t}$ for all $t \in W$, and hence we conclude that $\phi_{|mL|}|_{X_t}$ is generically finite for a general $t \in T$. This implies that $L|_{X_t}$ is big for every t in a neighborhood of 0.

Consider now the subset $\Sigma = \{t \in T \mid L|_{X_t} \text{ is not nef}\}$. We consider the base locus $\text{Bs}(|mL|)$ of $|mL|$. Note that this locus is a Zariski closed subset of X . We observe that $\text{Bs}(|mL|) \cap X_t \neq \emptyset$ for all $t \in \Sigma$, since it clearly contains the locus where L is not nef. On the other hand, taking t very general so that $|mL|_{X_t}|$ is base point free, we have $\text{Bs}(|mL|_{X_t}|) = \emptyset$, and since we can also ensure that the restriction map $H^0(\mathcal{O}_X(mL)) \rightarrow H^0(\mathcal{O}_{X_t}(mL))$ is surjective, we conclude that $\text{Bs}(|mL|) \cap X_t = \emptyset$. This implies that Σ is a finite set, and hence $L|_{X_t}$ is also nef for every t in a neighborhood of 0. $\square$

4. INVARIANCE OF PLURIGENERA FOR CANONICAL PAIRS

The following result is due to Hacon and McKernan (see [HM08] for a more general statement).

Theorem 4.1. *Let S be a normal divisor on a normal variety X , and let $\pi: X \rightarrow Z$ be a projective fibration to a normal affine variety Z . Let L be a Cartier divisor on X , and suppose that $L \sim_{\mathbb{Q}} k(K_X + S + B)$ for some positive integer k , where $(X, S + B)$ is purely log terminal and (S, B_S) is canonical, where B_S is defined by $K_S + B_S = (K_X + S + B)|_S$. Assume that $B \sim_{\mathbb{Q}} A + C$ where A is ample and C is effective, and that $S \not\subseteq \text{Supp}(C)$ and no component of $\text{Supp}(B_S)$ is contained in $\text{SBs}(K_X + S + B)$.*

Then $H^0(X, L) \rightarrow H^0(S, L|_S)$ is surjective.

In general, it is not possible to drop the assumption that no component of $\text{Supp}(B_S)$ is contained in $\text{SBs}(K_X + S + B)$. However Theorem 4.6 below gives an interesting example when this assumption may be dropped.

We begin with the following preliminary result (see [KM92, (12.3)] for a related statement).

Theorem 4.2. *Let $f: X \rightarrow T$ be a flat projective morphism of normal varieties where T is an affine curve. Assume that for some $0 \in T$ the fiber X_0 is normal and $\mathbb{Q}$ -factorial,*

$N^1(X/T) \rightarrow N^1(X_0)$ is surjective, and D is an effective $\mathbb{Q}$ -divisor whose support does not contain X_0 such that $(X_0, D|_{X_0})$ is a Kawamata log terminal pair with canonical singularities. Let $\psi: X \rightarrow Z$ be the contraction over T of a $(K_X + D)$ -negative extremal ray of $\text{NE}(X, T)$, and let $Z_0 = \psi(X_0)$.

If $\psi_0 := \psi|_{X_0}: X_0 \rightarrow Z_0$ is not an isomorphism, then it is the contraction of a $(K_{X_0} + D|_{X_0})$ -negative extremal ray, and:

- (a) If ψ is of fiber type, then so is ψ_0 .
- (b) If ψ is a divisorial contraction of a divisor G , then ψ_0 is a divisorial contraction of $G|_{X_0}$ (in particular $G|_{X_0}$ is irreducible), and $N^1(Z/T) \rightarrow N^1(Z_0)$ is surjective.
- (c) Assume that either $\text{SBs}(K_X + D/Z)$ does not contain any component of $\text{Supp}(D|_{X_0})$ or $D|_{X_0} - aK_{X_0}$ is nef over Z_0 for some $a > -1$. If ψ is a flipping contraction and $\psi^+: X^+ \rightarrow Z$ is the flip, then ψ_0 is a flipping contraction and, denoting X_0^+ the proper transform of X_0 on X^+ , the induced morphism $\psi_0^+: X_0^+ \rightarrow Z_0$ is the flip of $\psi_0: X_0 \rightarrow Z_0$. Moreover $N^1(X^+/T) \rightarrow N^1(X_0^+)$ is surjective.

Proof. We will repeatedly use the fact that T is affine so that for example a divisor is ample over T if and only if it is ample. For short, for a divisor L on X not containing X_0 in its support, we denote $L_0 := L|_{X_0}$.

If ψ is of fiber type, then it follows by the semicontinuity of fiber dimension that ψ_0 is also of fiber type. We can henceforth assume that ψ is birational.

Note that as $N^1(X/T) \rightarrow N^1(X_0)$ is surjective, any $(K_X + D)$ -negative extremal ray over T is also a $(K_{X_0} + D_0)$ -negative extremal ray. There is an ample $\mathbb{Q}$ -divisor H on X such that the contraction ψ is defined by $|m(K_X + D + H)|$ for any sufficiently divisible $m \geq 1$. Since $K_X + D + H$ is nef and big, the restriction map

$$H^0(m(K_X + D + H)) \rightarrow H^0(m(K_{X_0} + D_0 + H_0))$$

is surjective if m is sufficiently divisible by Kawamata–Viehweg’s vanishing theorem. This implies that ψ_0 coincides with the map defined by $|m(K_{X_0} + D_0 + H_0)|$ for all sufficiently divisible m , and thus it is the extremal contraction of the ray in question. In particular, ψ_0 has connected fibers and Z_0 is normal.

Assume that ψ is a divisorial contraction and let G be the (irreducible) divisor contracted by ψ . Then G dominates T and as $\psi(G)$ is irreducible of dimension $\leq \dim Z - 2$, all components of G_0 are contracted. It follows from the semicontinuity of fiber dimension and the injectivity of $N_1(X_0) \rightarrow N_1(X/T)$ that $\psi_0: X_0 \rightarrow Z_0$ is an extremal divisorial contraction of G_0 , and in particular G_0 is irreducible.

Assume that ψ is a flipping contraction, and suppose by way of contradiction that ψ_0 is a divisorial contraction so that Z_0 is $\mathbb{Q}$ -factorial. Note that Z_0 is canonical (if $\text{SBs}(K_X + D/Z)$ does not contain any component of D_0 , then in fact $(Z_0, (\psi_0)_*D_0)$ is canonical, whereas if $D|_{X_0} - aK_{X_0}$ is nef over Z , then $-K_{X_0} = \frac{1}{a+1}(-(K_{X_0} + D|_{X_0}) + (D|_{X_0} - aK_{X_0}))$ is ample over Z so that Z_0 is canonical). By Proposition 3.6, $K_Z + \psi_*D + Z_0$ is also $\mathbb{Q}$ -Cartier, and thus $K_Z + \psi_*D$ is $\mathbb{Q}$ -Cartier, since Z_0 is Cartier. This is a contradiction as ψ is a flipping contraction so that $K_Z + \psi_*D$ cannot be $\mathbb{Q}$ -Cartier. Therefore ψ_0 is a flipping contraction.

Note that $X^+ = \text{Proj}_Z \oplus_{m \geq 0} \psi_* \mathcal{O}_X(m(K_X + D))$. By Theorem 4.1, one sees that if Z is affine, then $\psi_* \mathcal{O}_X(m(K_X + D)) \rightarrow (\psi_0)_* \mathcal{O}_{X_0}(m(K_{X_0} + D_0))$ is surjective for all $m > 0$ sufficiently divisible.

We conclude that $X_0^+ \cong \text{Proj}_{Z_0} \oplus_{m \geq 0} (\psi_0)_* \mathcal{O}_{X_0}(m(K_{X_0} + D_0))$ so that $\psi_0^+ := \psi^+|_{X_0^+}$ is the flip of ψ_0 . The surjectivity $N^1(X^+/T) \rightarrow N^1(X_0^+)$ follows easily. $\square$

Corollary 4.3. *With the assumptions of Proposition 4.2, if $\psi|_{X_0}$ is a divisorial contraction, then its restriction $\psi|_{X_\eta}$ to the generic fiber of f is a divisorial contraction.*

Remark 4.4. As it will follow from the discussion in the next section, if $f: X \rightarrow T$ is a family of terminal $\mathbb{Q}$ -factorial Fano varieties then one can always reduce by finite base change to a setting where $N^1(X/T) \rightarrow N^1(X_0)$ is surjective. Note, moreover, that for a family of Fano varieties $D - aK_X$ is ample over T for every $a \gg 0$.

Remark 4.5. The necessity of the additional hypotheses in case (c) is manifested in the following example. Let $X \rightarrow T = \mathbb{A}^1$ be a flat family of smooth 2-dimensional quadrics degenerating to an $\mathbb{F}_2 = \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-1))$, and let E be the (-2) -curve on the central fiber $X_0 = \mathbb{F}_2$ (note that $K_X \cdot E = 0$). A line in one of the two rulings on the general fiber X_t sweeps out, under deformation over T , a divisor D on X such that $D|_{X_0} = E + F$, where F is a fiber of the projection $\mathbb{F}^2 \rightarrow \mathbb{P}^1$. The small contraction $X \rightarrow Z$ of E is $(K_X + \epsilon D)$ -negative for $\epsilon > 0$, and restricts to a divisorial contraction of X_0 .

Consider now a flat fibration of normal varieties $f: X \rightarrow T$ where $T = \text{Spec } R$ is an affine curve. Assume that X_0 is normal and $\mathbb{Q}$ -factorial and D is an effective $\mathbb{Q}$ -divisor whose support does not contain X_0 , such that $(X_0, D|_{X_0})$ is a Kawamata log terminal pair with canonical singularities and $K_X + D$ is big over T (similar results also hold if D is big over T , but we do not pursue it here). We fix $n \geq 1$ sufficiently divisible so that $n(K_X + D)$ is Cartier and the ring $R(n(K_X + D))$ is generated in degree one over R (the existence of such an n follows from [BCHM06]). Let

$$B := \frac{1}{n}(\text{Fix } |n(K_X + D)| \wedge nD) \text{ and } A := D - B,$$

where for two divisors L_1 and L_2 , the symbol $L_1 \wedge L_2$ denotes the largest divisor $\leq L_i$ for $i = 1, 2$.

Theorem 4.6. *With the above notation, assume moreover that X_0 has terminal singularities, $N^1(X/T) \rightarrow N^1(X_0)$ is surjective, and $A|_{X_0} - aK_{X_0}$ is ample for some $a > -1$.*

Then after possibly shrinking T near 0 we have that

$$H^0(m(K_X + D)) \rightarrow H^0(m(K_{X_0} + D|_{X_0}))$$

is surjective for all $m > 0$ sufficiently divisible.

Proof. We have seen that (X, D) is canonical and $\mathbb{Q}$ -factorial in a neighborhood of X_0 (see Corollary 3.2 and Proposition 3.6). After possibly shrinking T around 0, we can assume that (X, D) is everywhere canonical and $\mathbb{Q}$ -factorial. Notice that this reduction does not affect the assumption that $N^1(X/T) \rightarrow N^1(X_0)$ be surjective.

Given any rational number $0 < t \leq 1$, and letting $m \geq 1$ be sufficiently divisible (in particular we assume that m is a multiple of n and of the denominator of t), we can write $m(K_X + D) = m(K_X + A + tB) + m(1 - t)B$. Since $mB \leq \text{Fix}(m(K_X + D))$, it follows that $mtB \leq \text{Fix}(m(K_X + A + tB))$, and hence $\text{Supp } B \subseteq \text{SBs}(K_X + A + tB)$.

By Lemma 4.7 we have that for every rational number $0 < t \leq 1$ and all sufficiently divisible $m \geq 1$ the map

$$H^0(m(K_X + A + tB)) \rightarrow H^0(m(K_X + D))$$

induced by multiplication of a global section of $\mathcal{O}_X(m(1-t)B)$ is an isomorphism (here we are assuming that m is divisible enough so that $m(1-t)B$ is Cartier). Note in particular that $K_X + A + tB$ is big; we will use this fact towards the end of the proof.

We now fix a sufficiently small rational number $\epsilon > 0$ and run a $(X, A + \epsilon B)$ -minimal model program over T with scaling of $A - aK_X$. Let $\psi: X \dashrightarrow Z$ be the resulting map. Note that ψ contracts all the components of B . By the finiteness of minimal models [BCHM06], we can assume that ψ is a minimal model program for $(X, A + tB)$ for all $0 < t \leq \epsilon$. In particular, we have

$$H^0(m(K_X + A + tB)) \cong H^0(m(K_Z + \psi_*(A + tB)))$$

for any such t . Note that each contraction is $(A - aK_X)$ -positive and $(K_X + A)$ -semi-negative. Writing $(1 + a)K_X = (K_X + A) - (A - aK_X)$, we see that each contraction is K_X -negative. It follows that X remains canonical and hence by Theorem 4.2, each divisorial/flipping contraction on X corresponds to a divisorial/flipping contraction on X_0 . In particular ψ_0 extracts no divisors. Note that by what we have proven above² we have

$$K_{Z_0} + (\psi_0)_*(A_0 + tB_0) = (K_Z + (\psi)_*(A + tB))|_{Z_0}.$$

Since this divisor is nef for $0 < t \ll 1$, it follows by the negativity lemma (cf. [KM98, Lemma 3.39]) that $a_Q(X_0, A_0 + tB_0) \leq a_Q(Z_0, (\psi_0)_*(A_0 + tB_0))$, for any prime divisor Q over X_0 . Therefore

$$H^0(m(K_{X_0} + A_0 + tB_0)) \cong H^0(m(K_{Z_0} + (\psi_0)_*(A_0 + tB_0))).$$

As ψ contracts the support of B , one sees that ψ_0 contracts the support of B_0 . We deduce that $\text{Supp}(B_0) \subseteq \text{SBs}(K_{X_0} + A_0 + tB_0)$ for $0 < t \ll 1$, and therefore for all sufficiently divisible $m \geq 1$ the map

$$H^0(m(K_{X_0} + A_0 + tB_0)) \rightarrow H^0(m(K_{X_0} + D_0))$$

induced by multiplication of the restriction of the global section of $\mathcal{O}_X(m(1-t)B)$ is an isomorphism by Lemma 4.7.

Since $K_Z + \psi_*(A + tB)$ is nef and big, the restriction map

$$H^0(m(K_Z + \psi_*(A + tB))) \rightarrow H^0(m(K_{Z_0} + (\psi_0)_*(A_0 + tB_0)))$$

is surjective by Kawamata–Viehweg’s vanishing theorem. We obtain a commutative diagram

$$\begin{array}{ccccc} H^0(m(K_Z + \psi_*(A + tB))) & \longrightarrow & H^0(m(K_X + A + tB)) & \longrightarrow & H^0(m(K_X + D)) \\ \downarrow & & & & \downarrow \\ H^0(m(K_{Z_0} + (\psi_0)_*(A_0 + tB_0))) & \longrightarrow & H^0(m(K_{X_0} + A_0 + tB_0)) & \longrightarrow & H^0(m(K_{X_0} + D_0)) \end{array}$$

where the horizontal maps are isomorphisms and the first vertical map is surjective. It follows that the second vertical arrow is surjective. $\square$

Lemma 4.7. *Let (X, F) be a Kawamata log terminal pair. If $K_X + F$ is big, $\phi: X \dashrightarrow Y$ is a minimal model for (X, F) , and G is an effective $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor such that $\text{Supp } G \subset \text{SBs}(K_X + F)$, then $H^0(m(K_X + F)) \cong H^0(m(K_X + F + G))$ for all sufficiently divisible $m \geq 1$.*

²Say where “above”.

Proof. Note that $\phi_*G = 0$ and that for all rational numbers $t \geq 0$, we have $\text{Supp } G \subset \text{SBs}(K_X + F + tG)$. Let

$$\tau = \sup\{t \in [0, 1] \mid a_P(X, F + tG) \leq a_P(Y, \phi_*F) \text{ for every prime divisor } P \subset X\}.$$

One sees that $a_P(X, F) < a_P(Y, \phi_*F)$ if $\phi_*P = 0$, so that $\tau > 0$. If $\tau = 1$ the claim is clear. Otherwise, suppose that $\tau < 1$, and let $p: W \rightarrow X$ and $q: W \rightarrow Y$ be a common log resolution. We have

$$p^*(K_X + F + \tau G) = q^*(K_Y + \phi_*F) + E,$$

and $p_*E \geq 0$, so that by the negativity lemma applied along p , we have $E \geq 0$. By definition of τ , there is a component G' of G not contained in p_*E . Since $K_Y + \phi_*F$ is semiample, we have that $G' \not\subset \text{SBs}(K_X + F + \tau G)$. This is the required contradiction. $\square$

Theorem 4.6 can be made more precise, as follows.

Theorem 4.8. *With the assumptions of Theorem 4.6, after possibly shrinking T near 0 the map $H^0(k(K_X + D)) \rightarrow H^0(k(K_{X_0} + D|_{X_0}))$ is surjective for every positive integer k such that $k(K_X + D)$ is integral.*

Proof. After possibly shrinking T around 0, we can assume that (X, D) is everywhere canonical and $\mathbb{Q}$ -factorial. Notice that this reduction does not affect the assumption that $N^1(X/T) \rightarrow N^1(X_0)$ be surjective. Let $\mu: Y \rightarrow X$ be a log resolution of $(X, X_0 + D)$ and write

$$K_Y + Y_0 + P = \mu^*(K_X + X_0 + D) + E$$

where P and E are effective with no common components with $\mu_*P = D$ and $\mu_*E = 0$. Since (X, D) is canonical, we in fact have $P = (\mu^{-1})_*D$. Note that both kP and kE are integral divisors.

Recall that if N is an integral $\mathbb{Q}$ -Cartier divisor on X and F is an effective and exceptional divisor on Y , then $\mu_*\mathcal{O}_Y(f^*N + F) = \mathcal{O}_X(N)$. In particular we have

$$\mu_*\mathcal{O}_Y(k(K_Y + Y_0 + P)) = \mathcal{O}_X(k(K_X + X_0 + D)).$$

Therefore, it suffices to show that if $M = k(K_Y + Y_0 + P)$, then $H^0(Y, M) \rightarrow H^0(Y_0, M|_{Y_0})$ is surjective.

Since $K_X + X_0 + D \sim_{\mathbb{Q}} A + B$ where A is ample and B is effective and does not contain X_0 in its support, one sees that $M \sim_{\mathbb{Q}} C + G$ where C is ample and G is effective and does not contain Y_0 in its support. Pick any divisor

$$\Sigma \in |k(K_{Y_0} + P|_{Y_0})| = |M|_{Y_0}.$$

Since, by Theorem 4.4, $mk(K_X + X_0 + D)$ is Cartier and $H^0(mk(K_Y + Y_0 + P)) \rightarrow H^0(mk(K_{Y_0} + P|_{X_0}))$ is surjective for any m sufficiently divisible, we can pick a divisor

$$\Gamma \in |mk(K_Y + Y_0 + P)| = |mM|$$

such that

$$\Gamma|_{Y_0} = m\Sigma.$$

Then, if $\Theta = \frac{k-1-\epsilon}{mk}\Gamma + P + \frac{\epsilon}{k}G$ for some rational number $0 < \epsilon \ll 1$, we have

$$M - \Theta \sim_{\mathbb{Q}} \frac{1+\epsilon}{k}M - P - \frac{\epsilon}{k}G \sim_{\mathbb{Q}} K_Y + Y_0 + \frac{\epsilon}{k}C$$

so that by Nadel vanishing $H^1(Y, \mathcal{O}_Y(M - Y_0) \otimes \mathcal{J}(Y, \Theta)) = 0$. From the short exact sequence

$$0 \rightarrow \mathcal{O}_Y(M - Y_0) \otimes \mathcal{J}(Y, \Theta) \rightarrow \mathcal{O}_Y(M) \otimes \text{adj}(Y, Y_0; \Theta) \rightarrow \mathcal{O}_{Y_0}(M) \otimes \mathcal{J}(Y_0, \Theta|_{Y_0}) \rightarrow 0,$$

one sees that it suffices to show that any section of $\mathcal{O}_{Y_0}(M)$ defining Σ vanishes along $\mathcal{J}(Y_0, \Theta|_{Y_0})$. To see this notice that

$$\Theta|_{Y_0} - \Sigma \leq P|_{Y_0} + \frac{\epsilon}{k}G|_{Y_0}.$$

Since $(Y_0, P|_{Y_0})$ is Kawamata log terminal, then so is $(Y_0, P|_{Y_0} + \frac{\epsilon}{k}G|_{Y_0})$ for $0 < \epsilon \ll 1$. The claim now follows from [HM07, Lemma 5.4.2]. $\square$

5. NEF VALUES AND MORI CONES IN FAMILIES

Given $\mathbb{Q}$ -Cartier divisors A and B on a normal variety X , with A ample, the *nef value* (or *nef threshold*) of B with respect to A is defined by

$$\tau_A(B) := \min\{\lambda \in \mathbb{R} \mid B + \lambda A \text{ is nef}\}.$$

This invariant is of particular interest when $B = K_X + D$ for some Kawamata log terminal pair (X, D) such that $K_X + D$ is not nef. In this case, it follows from the Cone Theorem that $\tau = \tau_A(K_X + D)$ is rational and $K_X + D + \tau A$ is semiample.

The following is our main result.

Theorem 5.1. *Let $f: X \rightarrow T$ be a flat projective fibration of normal varieties where T is an affine curve. For some $0 \in T$, assume that the fiber X_0 over 0 is a normal $\mathbb{Q}$ -factorial variety, $N^1(X/T) \rightarrow N^1(X_0)$ is surjective, and D is an effective $\mathbb{Q}$ -divisor on X whose support does not contain X_0 such that $(X_0, D|_{X_0})$ is a Kawamata log terminal pair with canonical singularities. Let $\psi: X \rightarrow Z$ be the contraction of a $(K_X + D)$ -negative extremal ray, and let $Z_0 = \psi(X_0)$. Assume that $\psi_0 := \psi|_{X_0}: X_0 \rightarrow Z_0$ is not an isomorphism and either*

- (a) ψ_0 is small or of fiber type, or
- (b) ψ_0 is divisorial and either $\text{SBs}(K_X + D/Z)$ does not contain any component of $\text{Supp}(D|_{X_0})$ or $D|_{X_0} - aK_{X_0}$ is nef for some $a > -1$.

Then the restriction $\psi|_{X_\eta}$ to the generic fiber of f is not an isomorphism.

Proof. The cases in which ψ_0 is divisorial or of fiber type follow from Proposition 4.2 and Corollary 4.3. We may therefore assume that ψ_0 is a flipping contraction. We proceed by way of contradiction, and assume that $\psi|_{X_\eta}$ is an isomorphism. After shrinking T , we may assume that ψ is an isomorphism on the complement of X_0 . Moreover, after cutting down by the right number of general hyperplane sections of Z , we can assume without loss of generality that $\dim \psi(\text{Ex}(\psi)) = 0$.

We fix an ample divisor H on Z and an integer $k \gg 1$. Since ψ is small, we can assume that $\text{SBs}(K_X + D + k\psi^*H)$ does not contain any components of $\text{Supp}(D|_{X_0})$. Fix an integer $r \geq 1$ such that $r(K_X + D)$ is Cartier. For $t \neq 0$ and $m \gg 0$, we have

$$\chi(\mathcal{O}_{X_t}(mr(K_X + D + k\psi^*H))) = h^0(\mathcal{O}_{X_t}(mr(K_X + D + k\psi^*H))),$$

and since both sides are constant with respect to $t \in T$ (cf. Theorem 4.1), this implies that

$$(1) \quad \chi^+(\mathcal{O}_{X_0}(mr(K_X + D + k\psi^*H))) = 0 \quad \text{for } m \gg 0,$$

where for a sheaf $\mathcal{F}$, we denote $\chi^+(\mathcal{F}) := \chi(\mathcal{F}) - h^0(\mathcal{F})$.

For short, denote

$$Y = X_0 \quad \text{and} \quad L = r(K_X + D + \psi^*H)|_{X_0}.$$

Let $\psi_0^+ : Y^+ \rightarrow Z_0$ be the flip of $\psi_0 : Y \rightarrow Z_0$, and let L^+ be the proper transform of L on Y^+ . By possibly replacing k and r with multiples, we can assume without loss of generality that L^+ is very ample.

Let $n = \dim Y$. Fix $n - 1$ general divisors $B_j^+ \in |L^+|$, let $B_j \in |L|$ be their proper transform, and define

$$Y_i = B_1 \cap \cdots \cap B_i \quad \text{and} \quad Y_i^+ = B_1^+ \cap \cdots \cap B_i^+$$

for $i = 1, \dots, n - 1$. Note that each Y_i contains the exceptional locus $E = \text{Ex}(\psi_0)$, whereas Y_i^+ intersects $E^+ = \text{Ex}(\psi_0^+)$ properly. In particular, since E^+ has codimension ≥ 2 in Y^+ , we see that $Y_{n-1}^+ \cap E^+ = \emptyset$.

Fix any $1 \leq i \leq n - 1$. The birational map $Y \dashrightarrow Y^+$ induces an isomorphism

$$Y_i \setminus E \cong Y_i^+ \setminus E^+,$$

and since Y_i^+ is normal and $Y_i^+ \cap E^+$ has codimension ≥ 2 in Y_i^+ , every section of $\mathcal{O}_{Y_i^+ \setminus E^+}(mL^+)$ extends to a section of $\mathcal{O}_{Y_i^+}(mL^+)$. Therefore there is a natural map $H^0(\mathcal{O}_{Y_i}(mL)) \rightarrow H^0(\mathcal{O}_{Y_i^+}(mL^+))$, and we obtain a commutative diagram

$$\begin{array}{ccc} H^0(\mathcal{O}_Y(mL)) & \longrightarrow & H^0(\mathcal{O}_{Y^+}(mL^+)) \\ \downarrow & & \downarrow \\ H^0(\mathcal{O}_{Y_i}(mL)) & \longrightarrow & H^0(\mathcal{O}_{Y_i^+}(mL^+)). \end{array}$$

As Y and Y^+ are isomorphic in codimension 1, the top horizontal arrow is an isomorphism. For $m \gg 0$, the map $H^0(\mathcal{O}_{Y^+}(mL^+)) \rightarrow H^0(\mathcal{O}_{Y_i^+}(mL^+))$ is surjective. Therefore, for any section $s \in H^0(\mathcal{O}_{Y_i}(mL))$ there is a section $\tilde{s} \in H^0(\mathcal{O}_Y(mL))$ whose restriction to $Y_i \setminus E$ coincides with the restriction of s . On the other hand, if $m \gg 0$, then any two sections of $\mathcal{O}_{Y_i}(mL)$ that agree on the open set $Y_i \setminus E$ are necessarily the same section. This follows from the fact that we can fix a sufficiently large integer k so that the sheaf $\mathcal{O}_{Y_i}/I_E^k$ is supported on the closure of $Y_i \setminus E$, and we have $H^0(\mathcal{O}_{Y_i}(mL)) = H^0(\mathcal{O}_{Y_i}(mL) \otimes I_E^k)$ for $m \gg 0$.

In conclusion, we have shown that, for every $1 \leq i \leq n - 1$ and $m \gg 0$, the restriction map $H^0(\mathcal{O}_Y(mL)) \rightarrow H^0(\mathcal{O}_{Y_i}(mL))$ is surjective, and therefore that the map

$$(2) \quad H^0(\mathcal{O}_{Y_{i-1}}(mL)) \rightarrow H^0(\mathcal{O}_{Y_i}(mL))$$

is surjective (here $Y_0 = Y$). This implies that, for every $i = 0, \dots, n - 1$,

$$\chi^+(\mathcal{O}_{Y_i}(mL)) = 0 \quad \text{for } m \gg 0.$$

Indeed the case $i = 0$ is given by (1), and considering the exact sequence

$$0 \rightarrow \mathcal{O}_{Y_{i-1}}((m-1)L) \rightarrow \mathcal{O}_{Y_{i-1}}(mL) \rightarrow \mathcal{O}_{Y_i}(mL) \rightarrow 0$$

the assertion follows, via induction, by the surjectivity of the restriction map (2).

Therefore, taking $i = n - 1$, we have $\chi^+(\mathcal{O}_{Y_{n-1}}(mL)) = 0$ for $m \gg 0$. But this is impossible. Indeed we have $Y_{n-1} = C \sqcup F$, where C is the isomorphic image of Y_{n-1}^+ and hence disjoint from E , and F is a scheme supported exactly on E . Then, taking m sufficiently large, we have

$$\chi^+(\mathcal{O}_{Y_{n-1}}(mL)) = \chi^+(\mathcal{O}_C(mL)) + \chi^+(\mathcal{O}_F(mL)) = \chi(\mathcal{O}_F(mL)),$$

which is a polynomial in m of degree $d = \dim F > 0$ (cf. [Har77, Exercise III.5.2]). $\square$

We have the following immediate consequence.

Corollary 5.2. *With the assumptions as in Theorem 5.1 (including that either $\text{SBs}(K_X + D/Z)$ does not contain any component of $\text{Supp}(D|_{X_t})$ for any $t \in T$ or $D|_{X_0} - aK_{X_0}$ is nef for some $a > -1$), suppose that $K_{X_0} + D|_{X_0}$ is not nef over T .*

If A is an ample line bundle on X , then the nef values $\tau_{A|_{X_t}}(K_{X_t} + D|_{X_t})$ is independent of t .

Remark 5.3. When $f: X \rightarrow T$ is a smooth family and $D = 0$, this result was proven by Wiśniewski in [Wiś91] using the Hard Lefschetz Theorem and the fact that in this case f is topologically a locally trivial fibration.

Remark 5.4. As we have already discussed in Remark 4.5, the example of a family of quadric surfaces degenerating to an $\mathbb{F}_2$ shows that in general the additional assumptions on D required in case (b) of Theorem 5.1, and hence in the above corollary, cannot be dropped.

As an application of Corollary 5.2, we obtain the following result.

Theorem 5.5. *Let $f: X \rightarrow T$ be a flat projective family of Fano varieties with terminal $\mathbb{Q}$ -factorial singularities over a smooth curve T .*

Then the cone of effective curves $\text{NE}(X_t)$ is locally constant in the family.

Remark 5.6. The assertion that $\text{NE}(X_t)$ be locally constant under deformation and can be interpreted in the analytic category by saying that the Mori cones are mapped to each other via isomorphisms between the vector spaces $N_1(X_t)$ that are induced by small deformations (it is instructive to consider here the example of a family of smooth 2-dimensional quadrics degenerating to an $\mathbb{F}^2$, where the cone of curves of the central fiber, in spite of being abstractly isomorphic to that of any other fiber, is strictly larger than the one obtained by degeneration from a general fiber). However, it does not quite make sense in the Zariski topology, as in general there is no natural isomorphism $N_1(X_t) \rightarrow N_1(X_u)$ for $t \neq u$ in T unless one first fixes a path joining t to u . This is the case, for instance, for a quadric fibration $f: X \rightarrow T$ with all fibers isomorphic to a smooth quadric $\mathbb{P}^1 \times \mathbb{P}^1$ and relative Picard number $\rho(X/T) = 1$ (an explicit example is given by the family of quadrics of equation $\{xy + z^2 + tu^2 = 0\} \subset \mathbb{P}^3 \times \mathbb{C}^*$, where (x, y, z, u) are the homogeneous coordinates on $\mathbb{P}^3$ and $t \in \mathbb{C}^*$).

Remark 5.7. The example of a family of quadrics $\mathbb{P}^1 \times \mathbb{P}^1$ degenerating to an irreducible quadric cone shows that the theorem fails if one relaxes the assumptions on singularities from terminal to canonical. The example discussed in Remark 4.5 shows instead that one cannot hope for a similar result for families of log Fano varieties.

One can restate Theorem 5.5 more precisely, as follows. We know by Corollary 3.2 that that a flat projective family over a smooth curve $f: X \rightarrow T$ of Fano varieties with terminal $\mathbb{Q}$ -factorial singularities satisfies the conditions in [KM92, (12.2.1)]. We can therefore consider the local systems $\mathcal{GN}^1(X/T)$ and $\mathcal{GN}_1(X/T)$, and the sheaf of cones $\mathcal{GNE}(X/T)$, introduced in [KM92, Section 12] (notice that, these local systems, that were defined [KM92] using rational coefficients, are here considered with real coefficients). These are sheaves on T in the analytic topology. For any analytic open set $U \subseteq T$, these are given by:

$$\mathcal{GN}^1(X/T)(U) = \{ \text{sections of } N^1(X/T) \text{ over } U \text{ with open support} \},$$

where, in our situation, $\mathcal{N}^1(X/T)$ is the functor given by $N^1(X \times_T T'/T')$ for any $T' \rightarrow T$,

$$\mathcal{GN}_1(X/T)(U) = \left\{ \begin{array}{l} \text{flat families of 1-cycles } C/U \subseteq X \times_T U \text{ with real} \\ \text{coefficients, modulo fiberwise numerical equivalence} \end{array} \right\},$$

and

$$\mathcal{GN}\mathcal{E}(X/T)(U) = \left\{ \begin{array}{l} \text{flat families of 1-cycles } C/U \subseteq X \times_T U \text{ with nonnegative} \\ \text{coefficients, modulo fiberwise numerical equivalence} \end{array} \right\}.$$

It is shown in [KM92, (12.2)] that $\mathcal{GN}^1(X/T)$ and $\mathcal{GN}_1(X/T)$ are dual local systems with finite monodromy and, moreover, that $\mathcal{GN}^1(X/T)|_t = N^1(X_t)$, $\mathcal{GN}_1(X/T)|_t = N_1(X_t)$ and $\mathcal{GN}\mathcal{E}(X/T)|_t = \text{NE}(X_t)$ for very general $t \in T$. In fact, since in our setting we have $R^1 f_* \mathcal{O}_X = R^2 f_* \mathcal{O}_X = 0$, we can apply the same arguments in the proof of [KM92, (12.1.1)] to obtain the following property.

Proposition 5.8. *With the above assumptions, we have*

$$\mathcal{GN}^1(X/T)|_t = N^1(X_t) \quad \text{and} \quad \mathcal{GN}_1(X/T)|_t = N_1(X_t)$$

for all $t \in T$. In particular, the Picard number $\rho(X_t)$ is independent of $t \in T$.

Proof. We consider T as an analytic space, and work in the analytic topology over it. We fix an arbitrary point $t \in T$. After restricting, we momentarily assume that T is a small open disk around t .

We can find a subset $S \subset X$ of codimension ≥ 2 (in fact ≥ 3), whose restriction to X_t has codimension ≥ 2 as well, such that f restricts to a smooth morphism

$$f^\circ: X^\circ := X \setminus S \rightarrow T.$$

Then, as T is contractible and f° is topologically trivial, $R^2(f^\circ)_* \mathbb{Z}_{X^\circ}$ is a trivial local system with fiber isomorphic to $H^2(X_t^\circ, \mathbb{Z})$, where $X_t^\circ := (f^\circ)^{-1}(t)$. Since X is normal, for any open set $U \subseteq X$, every regular function on $U \cap X^\circ$ extends to a regular function on U , and thus $\mathcal{O}_{X^\circ}(U \cap X^\circ) = \mathcal{O}_X(U)$. By the definition of Čech cohomology, it follows that $R^i(f^\circ)_* \mathcal{O}_{X^\circ} \cong R^i f_* \mathcal{O}_X$, and similarly we have $H^i(\mathcal{O}_{X_t^\circ}) \cong H^i(\mathcal{O}_{X_t})$. On the other hand, we have $R^i f_* \mathcal{O}_X = 0$ and $H^i(\mathcal{O}_{X_t}) = 0$ for $i > 0$ by Kawamata–Viehweg vanishing, since f is a family of Fano varieties. Then, by pushing forward the exponential exact sequence of X° , we obtain a chain of isomorphisms

$$\text{Pic}(X^\circ) \cong H^0(R^1(f^\circ)_* \mathcal{O}_{X^\circ}^*) \cong H^0(R^2(f^\circ)_* \mathbb{Z}_{X^\circ}) \cong H^2(X_t^\circ, \mathbb{Z}) \cong \text{Pic}(X_t^\circ).$$

Note that we also have isomorphisms $\text{Pic}(X) \cong \text{Pic}(X^\circ)$ and $\text{Pic}(X_t) \cong \text{Pic}(X_t^\circ)$. Therefore we conclude that the restriction from X to X_t induces an isomorphism $\text{Pic}(X) \cong \text{Pic}(X_t)$.

Recall that we have been assuming that T is simply connected. In general, if T is an algebraic curve, then we only have that $\mathcal{GN}^1(X/T)|_t = N^1(X_t)$. The property for $\mathcal{GN}_1(X/T)$ follows by duality. $\square$

Using these local systems, Theorem 5.5 can be restated in the following way.

Theorem 5.9. *Let $f: X \rightarrow T$ be a flat projective family over a smooth curve of Fano varieties with terminal $\mathbb{Q}$ -factorial singularities.*

Then

$$\mathcal{GN}\mathcal{E}(X/T)|_t = \text{NE}(X_t)$$

for every $t \in T$. Moreover, for every $t, u \in T$ and every path γ from t to u there is a natural isomorphism $h: N_1(X_t) \rightarrow N_1(X_u)$ (depending on γ) such that $h(\text{NE}(X_t)) = \text{NE}(X_u)$.

Proof. After base change, we can assume without loss of generality that $\mathcal{GN}^1(X/T)$ has trivial monodromy, and thus $N^1(X/T) \rightarrow N^1(X_t)$ is surjective by Proposition 5.8.

The first assertion follows directly from Theorem 5.1. Indeed, after identifying $\mathcal{GN}_1(X/T)|_t = N_1(X_t)$, we have an inclusion $\mathcal{GNE}(X/T)|_t \subseteq \text{NE}(X_t)$ which is an equality since by Theorem 5.1 every extremal ray of $\text{NE}(X_t)$ is spanned by the class of a curve that deforms in a family dominating T .

For the second assertion we use Corollary 5.2. We can assume without loss of generality that t is a general point of T . After trivializing the local system $\mathcal{GN}^1(X/T)$, we have a natural isomorphism $g: N^1(X_t) \rightarrow N^1(X_u)$ (independent of any path). If $L_1, \dots, L_k$ are divisors on X such that their restrictions $L_i|_{X_t}$ are the good supporting divisors of the facets of $\text{NE}(X_t)$, then it follows by Corollary 5.2 that the divisors $L_i|_{X_u}$ are the good supporting divisors of the facets of $\text{NE}(X_u)$. The assertion follows by observing that $[L_i|_{X_u}] = g([L_i|_{X_t}])$. $\square$

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DEPARTMENT OF MATHEMATICS, UNIVERSITY OF UTAH, 155 SOUTH 1400 EAST, SALT LAKE CITY, UT 84112-0090, USA

E-mail address: `defernex@math.utah.edu`

DEPARTMENT OF MATHEMATICS, UNIVERSITY OF UTAH, 155 SOUTH 1400 EAST, SALT LAKE CITY, UT 84112-0090, USA

E-mail address: `hacon@math.utah.edu`